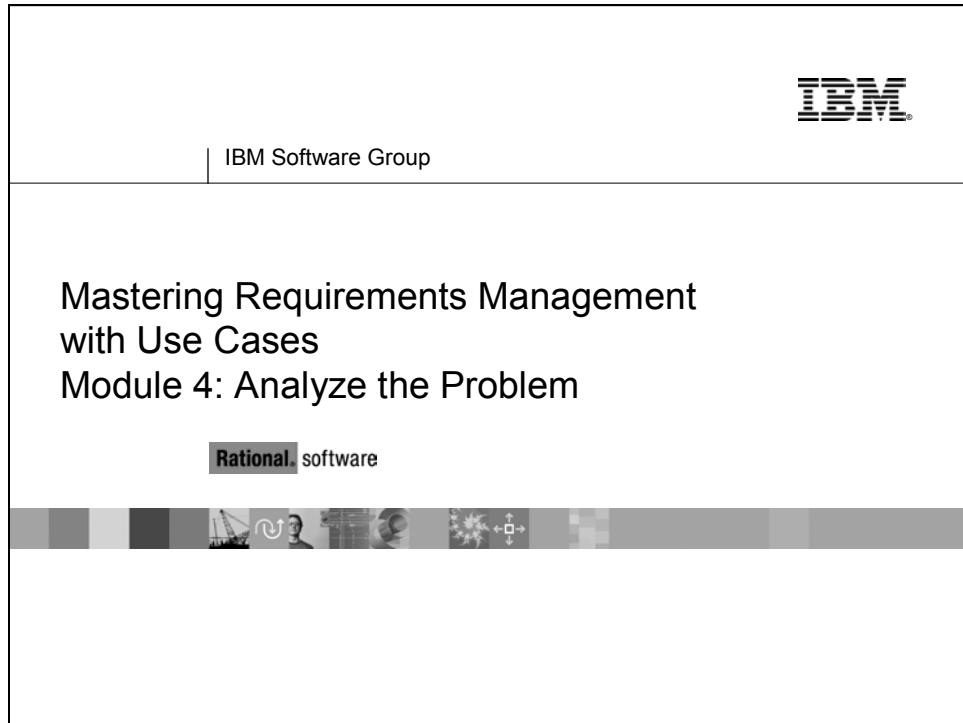


► ► ► **Module 4** **Analyze the Problem**



Topics

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Objectives

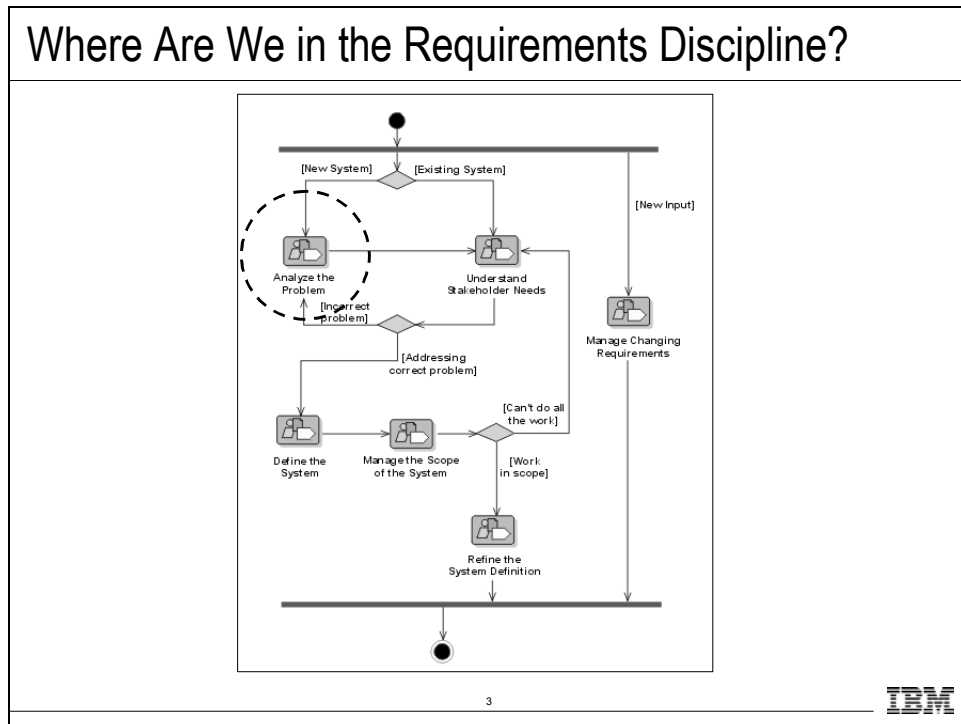
Objectives

- ♦ Define “problem analysis” and its goal.
- ♦ Describe activities for problem analysis.
 - Identify the stakeholders.
 - Gain agreement on the problem.
 - Find actors and define system boundaries.
 - Start development of the project Vision.
 - Define a problem statement.
 - Identify constraints on the project.
 - Establish common vocabulary.

2



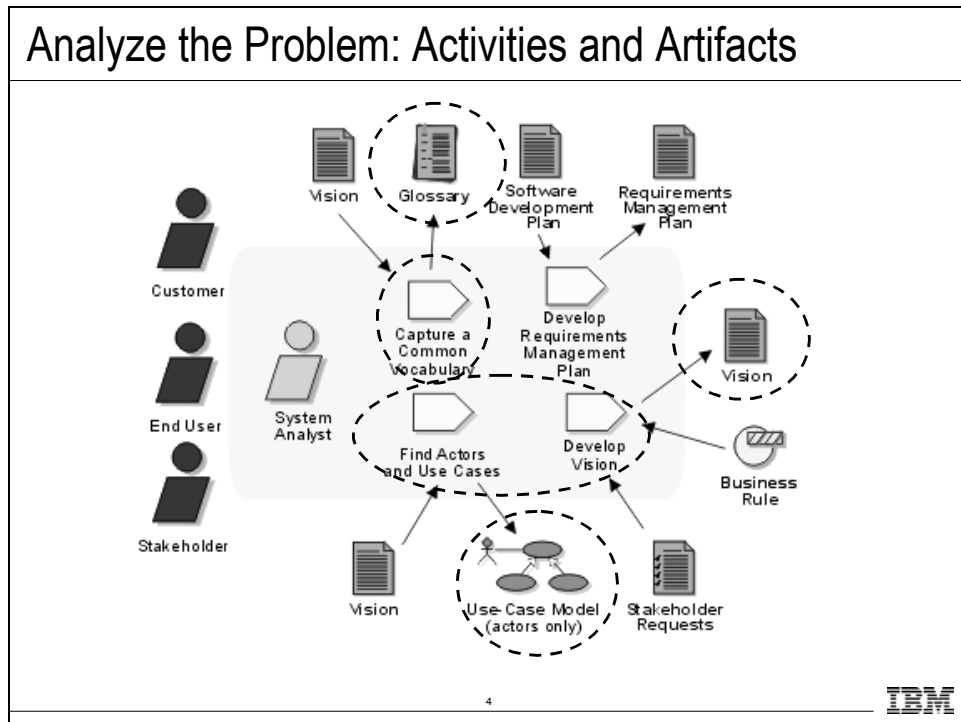
Where Are We in the Requirements Discipline?



Where does the Analyze the Problem detail workflow fit in our development process? This diagram shows the Requirements Discipline in the Rational Unified Process. In this module, we study the workflow detail called “Analyze the Problem.” It is a part of the Requirements Discipline that is performed when the problem to be solved is not already well-understood.

Where is the “Analyze the Problem” activity done in your own software development process?

Analyze the Problem: Activities and Artifacts



The first step in any problem analysis is to make sure that all parties involved agree on what the problem is that needs to be solved, or opportunity that will be realized by the system. In order to help avoid misunderstandings, it is important to agree on common terminology which will be used throughout the project. Starting early in the lifecycle, define your project terms in a glossary which can be maintained throughout the life of the project.

To fully understand the problem(s) that needs to be addressed, it is very important to know who the stakeholders are in the conceptual vision for the project. Note that some of these stakeholders, the users of the system, are represented by actors in your use-case model.

The primary artifact in which you capture the information gained from your problem analysis is the Vision, which identifies the high-level user or customer view of the system to be built. In the Vision, initial high-level requirements identify the key features it is desired that the appropriate solution will provide. These are typically expressed as a set of high-level features the system might possess in order to solve the most critical problems.

Key stakeholders should be involved in gathering the set of features to be considered, which might be gathered in a requirements workshop. The features can then be assigned attributes, such as rationale, relative value or priority, source of request and so on, so that dependencies and work plans can begin to be managed.

To determine the initial scope for your project, the boundaries of the system must be agreed upon. The system analyst identifies users and systems represented by actors which interact with the system.

Problem Analysis

Problem Analysis

- ♦ Is the process of understanding real-world problems, how they relate to stakeholder needs, and proposing solutions to meet those needs.
- ♦ What is the goal of problem analysis?
 - Gain a better understanding before development begins.
 - Identify root causes.
 - Help find the right solution.
 - Avoid the “Yes, but...”
 - Minimize extra work.



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What is the problem, *really*?

Step back and analyze what problem our system addresses.

There may be many different solutions that could fit our Problem Space. Our first step is to understand the problem, identify the best solution, and figure out the stakeholders' needs in this area.

Why is this important?

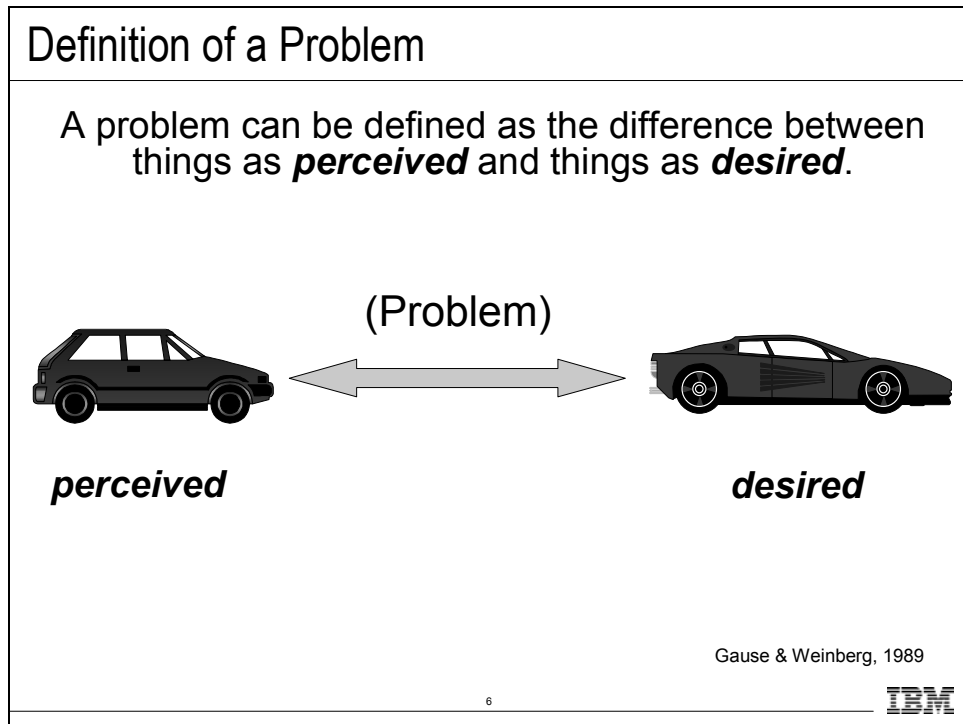
First, if you want to satisfy the customer's real needs, you must understand what problem they are trying to solve. You want to avoid hearing a “Yes, but...” when you deliver the final product, for example: “Yes, it meets the requirements, but it does not solve my problem.”

Second, if you want to avoid extra work, it pays to focus on the real problem and focus on the part of the problem that you actually need to solve. Solving the wrong part of the problem means you might have to go back and redo much of your work.

Third, understanding the problem is the first step in understanding the requirements. Customers often describe the problem in terms of their own needs, but each need can be associated with a high-level requirement:

- In the problem statement, the subject is the customer
“I need to ...”
- In the corresponding requirement, the subject is the system
“The system provides ...”

Definition of a Problem



If a difference does not exist between what you perceive you have and what you desire to have, then there is no problem. In our process, we first need to understand if there really is a problem.

This definition of a “problem” is different from many other ways to define the word “problem.” It addresses the gap between our perceptions of what our customer wants and what they actually desire. One advantage of this definition is that it emphasizes that there are many ways to solve a problem:

- Change the customers’ perception of what they have now. For example, many of the requests for enhancements that Microsoft gets are for features that already exist.
- Change the customers’ current desire. For example, if the customers find that what they want will cost millions of dollars over their budget, then maybe they will scale back their desires and focus on solving a simpler problem.
- Change the gap between what customers want and what they actually desire. A software solution is usually an attempt to narrow the gap.

Problem Analysis Steps

Problem Analysis Steps

- ◆ Identify stakeholders.
- ◆ Understand the root causes.
- ◆ Gain agreement on the problem definition.
- ◆ Identify constraints on the system or project.
- ◆ Identify and validate the solution against the root causes.
- ◆ Define the solution system boundary.

7



Above is a list of activities that you go through when performing a problem analysis. Start by identifying stakeholders for the problem. These stakeholders are people who are affected by the problem. Next, understand the root causes for the problem (the real problem behind the problem). Once the real problem (root cause) has been identified, get all your stakeholders to agree to a description of the problem.

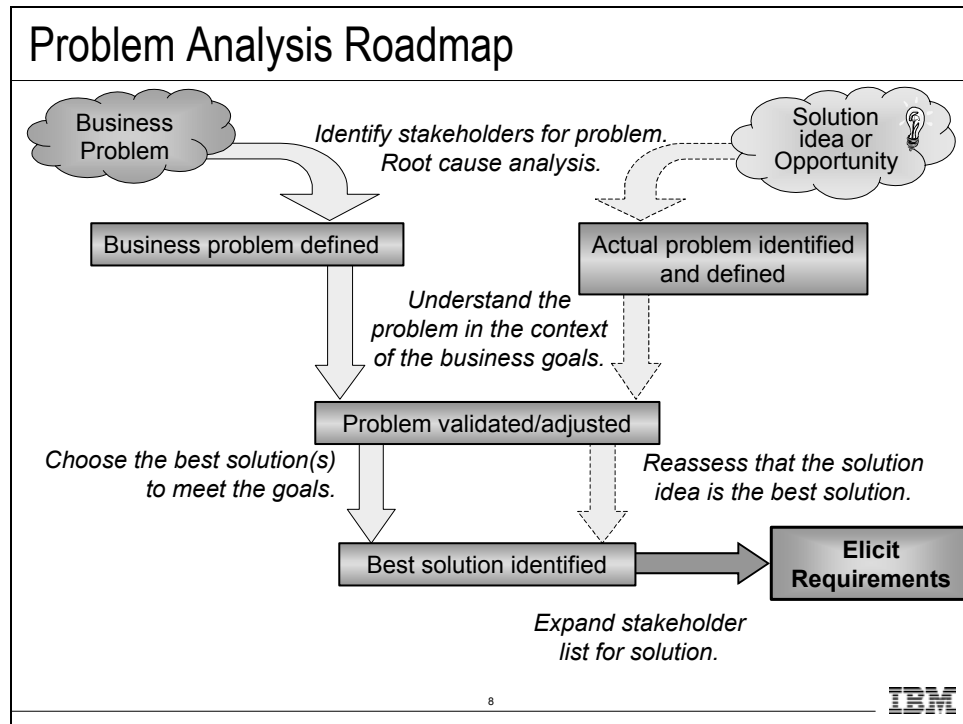
Armed with a description of the problem, identify any constraints that may be imposed on any solution. These constraints typically limit the solution you may provide.

Next, identify possible solutions for the problem. It is important that you identify more than one possible solution. By doing this, you can then compare and contrast the solutions to find the best possible solution.

An additional step not described above is to expand the set of stakeholders to include those affected by the specific solution you have identified.

The final step is to define the boundary of the solution. This is used to limit the scope during Understand Stakeholder's Needs.

Problem Analysis Roadmap



This slide shows the two typical approaches to defining a system. One approach is where someone has an idea or recognizes a need and develops a product to solve that problem. The other stems from the realization that there is some problem that needs solving. In the case of the former, an analysis of the business problem is often implied or thought out informally.

No matter which perspective you start from, it is important to understand where the system fits in the context of the goals of the organization. Failing to do this could mean you develop a system that adds little or no business value. In fact, it is possible that the wrong system could impede an organization from achieving its business goals.

Note: The Standish Group uses the term Business Objectives. The Rational Unified Process uses the term Business Goals. They are the same thing.

A common reason for project failure is that people start with a solution idea and head straight to development. When the project starts to fail, they almost always back track through this workflow.

Stakeholders: Definitions

Stakeholders: Definitions

- ◆ Stakeholder
 - An individual who is materially affected by the outcome of the system or the project(s) producing the system.
- ◆ Stakeholder Representative
 - A stakeholder representative represents one or more stakeholders. They are directly involved in the steering, shaping, and scoping of the project.

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There are many sources of requirements. Requirements may come from anyone with an interest in the outcome of a project. It is important to determine who those sources are early in your analysis. End users are the most obvious group to consider, but there are many others that should be included: for instance, who will maintain the system; who has an economic interest in the outcome; who will develop, test, and support the product. Failure to consider some of these stakeholders can lead to unexpected project costs and failures.

For example, you might release your project and have the Helpdesk manager saying: “The system is too complicated. We’re losing money on support because our customers cannot use the system.”

A Manager of Installations: “It takes too long to install, so we’re losing money with each installation; we can’t bill for the extra time.”

You have to consider the input of these stakeholders as part of the early evaluation process.

To keep the size of your stakeholder list manageable, it is important that you identify one or two people to represent each stakeholder type. The stakeholder representative should be a domain expert and have the authority to speak on behalf of the group they represent.

Some example stakeholders are:

Sponsors: Business managers, financiers, shareholders, champions, sellers, marketers, steering committee members, investors.

Authorities: Legislative authorities, standards organizations, organizational governance departments, external and internal regulatory bodies, domain experts, technology experts.

Identify the Stakeholders

Identify the Stakeholders

- ♦ Each group of stakeholders needs a representative.
- ♦ Not all stakeholder groups need to be consulted.
 - Some will provide requirements.
 - Customers, users, system administrators
 - Some may not provide requirements.
 - Shareholders

Who are some of the stakeholders for your projects?

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Part of analyzing the problem is understanding who has the problem. You need to understand the *real* stakeholders in your project, in addition to the customer who is paying the bill.

Make sure you consider all those affected by the outcome of the system, including stock shareholders, system maintainers, and developers. It is important to consider which stakeholders are sources of requirements. While we must identify all our stakeholders, it is important to focus your efforts where you receive the best return. Some of your stakeholders will not provide you with many or any requirements (for example, shareholders). While these interests certainly need to be considered and represented, you should focus your efforts accordingly. For example, if a system is mission critical, then a shareholder is certainly a stakeholder, but will they have any meaningful requests of the system? Generally, they will not.

Who are possible stakeholders for your own projects?

Describe Stakeholders in the Vision Document

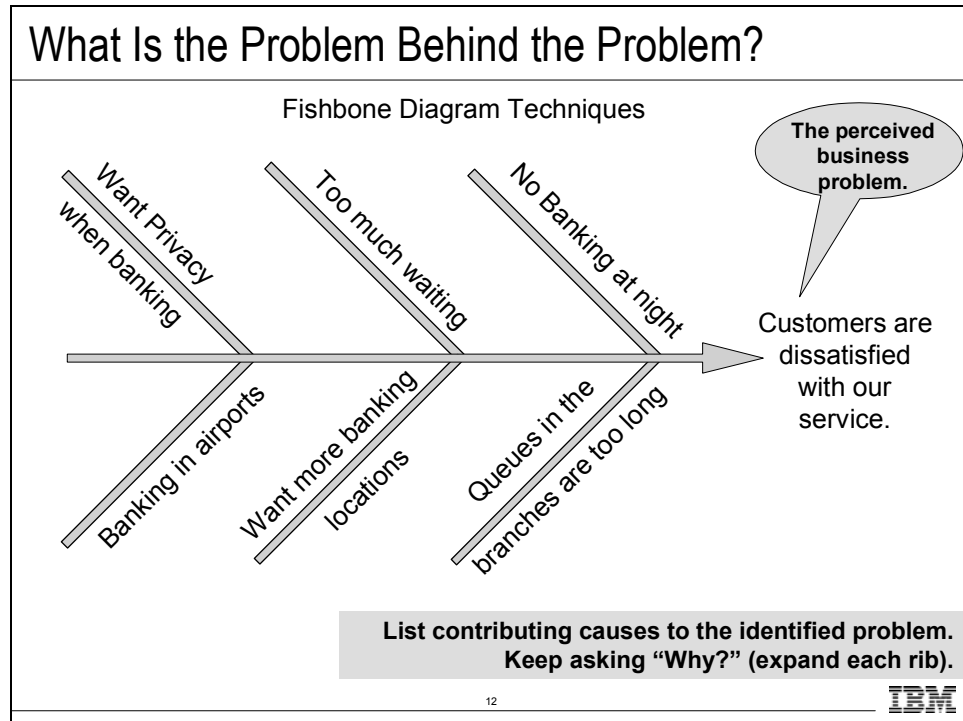
Describe Stakeholders in the Vision Document	
Stakeholder	Registrar
Representative	Kelly Hansen
Description	User
Type	The Registrar is typically a college-educated professional with full computer skills. The Registrar is trained and experienced with the use of the current batch-oriented registration .
Responsibilities	The Registrar is responsible for administering course registration for each school term. This includes supervising administrative and data entry personnel.
Success Criteria	The registrar's primary responsibility will be maintaining student and professor databases, and opening/closing courses to registration. The registrar's office will also be required to perform
Involvement	The registrar's primary responsibility will be maintaining student and professor databases, and opening/closing courses to registration. The registrar's office will also be required to perform.....
Deliverables	Management reviewer – especially related to functionality and usability of features required by the Registrar staff.
Comments/ Concerns	None

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The example above is an extract from the *Stakeholder Profiles* section of the Vision. This is an example of a very thorough profile. You may or may not wish to go into this level of detail when describing your stakeholders.

What Is the Problem Behind the Problem?



"Customers are dissatisfied with our service." Is this really the true problem?

The fishbone diagram is one method for finding the "root cause" of a problem.

The spines are contributing causes to the problem.

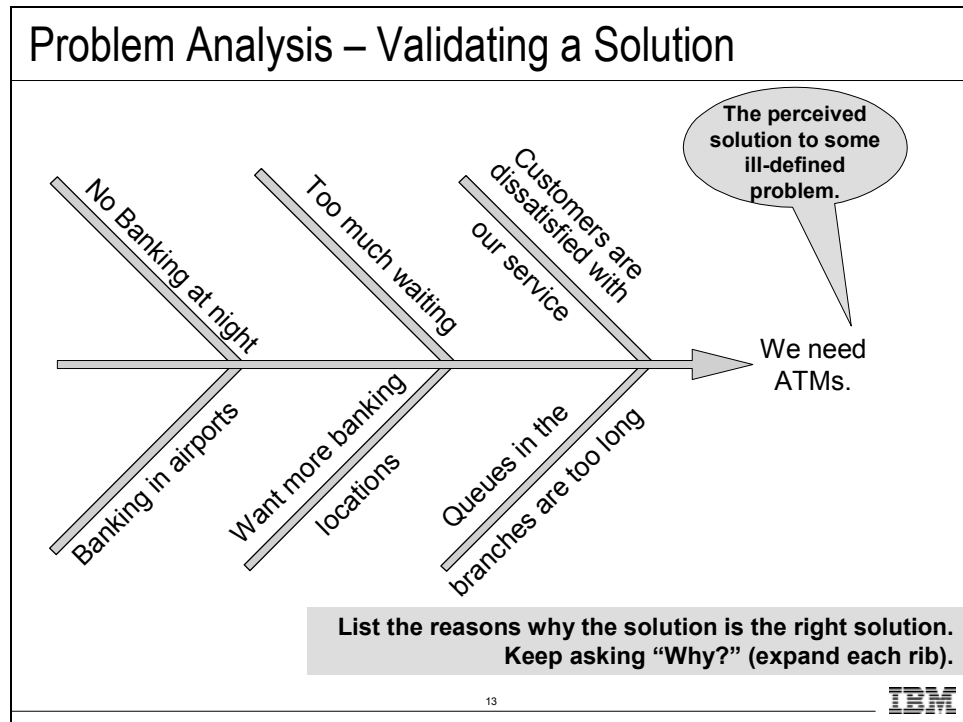
Once the root causes have been defined, focus on the ones that contribute most to the problem. You can then further explore the most important root causes to reveal contributing factors to these causes: expand each rib by building spines branching off the rib, or do a separate second-level fishbone diagram with a first-level root cause as "the problem."

What you often discover is that there may be more than one simple problem involved. Instead, there may be many dimensions to and perceptions of what the problem really is. You may need to help the customer focus on the dimensions of the problem that they most need to solve.

You help the customer focus on the problem that really needs to be solved.

Once you have a list of problems, you then need to perform a little analysis to identify the best solution to the biggest contributors to the problem. This step comes after you have all agreed to the main contributing problems.

Problem Analysis – Validating a Solution



This technique can also be used to validate a solution. Many of our customers come to us with a request for a solution before they understand what problem they are really solving. For example, "We need automatic teller machines." If we simply implement their solution without first analyzing the problem, we risk solving the wrong problem. After their own solution is delivered, the customer might find the delivered system useless for solving the real problem.

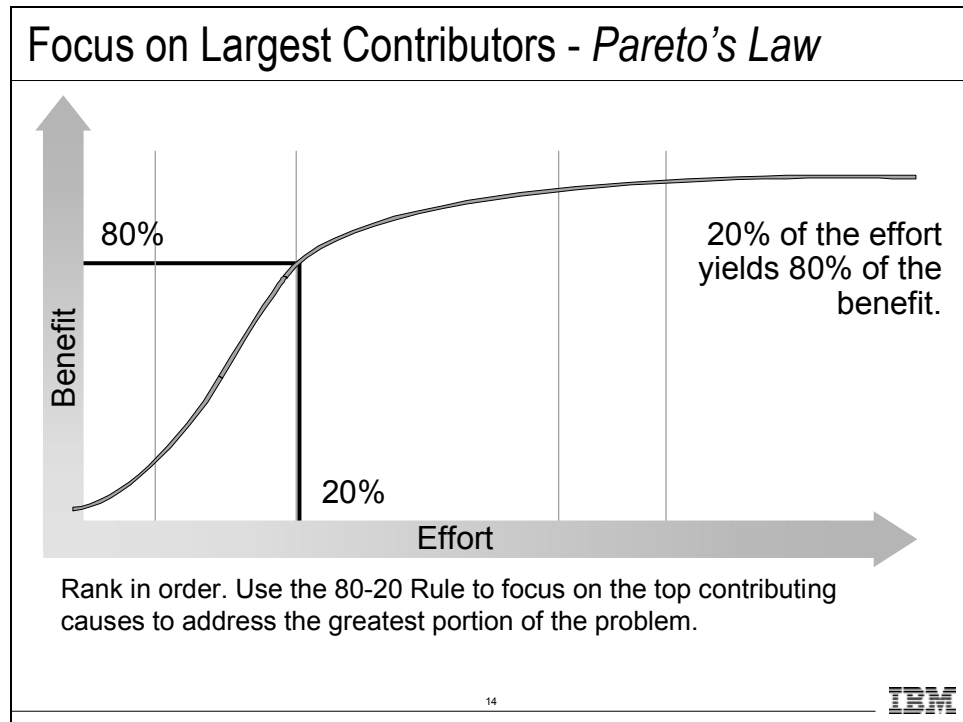
When you are presented with a solution like this, the customer has probably already done some informal problem analysis (usually in their head). The fishbone diagram can be used to validate that the solution really will solve the right problem.

Once the root causes have been defined, focus on the ones that contribute most to the problem. You can then further explore the most important root causes to reveal contributing factors to these causes: expand each rib by building spines branching off the rib, or do a separate second-level fishbone diagram with a first-level root cause as "the problem."

What you often discover is that there may be more than one simple problem involved. Instead, there may be many dimensions to and perceptions of what the problem really is. You may need to help the customer focus on the dimensions of the problem that they most need to solve.

Once you have a list of problems, look at the solution you are proposing and assess whether it is the best solution to the main contributing problems. This step comes after you have all agreed to the main contributing problems.

Focus on Largest Contributors - Pareto's Law

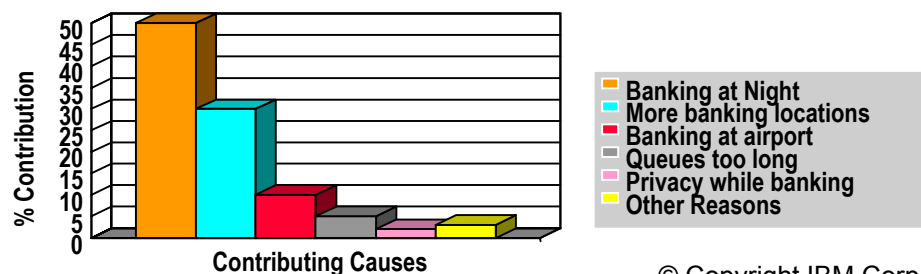


One way of focusing on the most important root causes is the Pareto Principle, or "80-20 rule." A Pareto chart ranks the contributing causes in the order of their contribution to the problem, so it is easy to see the largest contributors. A Pareto diagram can also be thought of as a "Fishbone diagram with numbers." Assign a weight to each bone and then graph it.

Notice that in the example, 80% of the problem is caused by the top two contributors. Where do the percentages come from? In some situations, you can measure how much each factor contributes to a problem. For example, if a mail order business is getting too many products returned, the business can ask each customer why the items were returned and tabulate the factors contributing to returns.

In other situations, the percentage contributions are estimates. For example, if a business is investigating the need for ATM machines, they can use focus groups to estimate the perceived problems associated with a lack of such facilities.

In many cases, rough estimates are used rather than hard data. In some cases, the top contributing causes are obvious. Focus on the top contributing causes rather than coming up with solutions to minor factors.



Understand the Broader Context of the Problem

Understand the Broader Context of the Problem

- ♦ A lack of understanding the business and its goals increases risk.
- ♦ Does the problem have an organization/process component?
- ♦ Does the team understand the domain in which the problem exists?
- ♦ Does solving the problem present the opportunity to make process improvements?

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Armed with a clear definition of the root causes to the business problem, it is important to ensure that the solution supports the business both operationally and strategically.

A new system might also provide the organization with the opportunity to improve the way it operates.

To do this, a more thorough understanding of the business may be required. Business modeling is a tool that you can use to help you answer these questions.

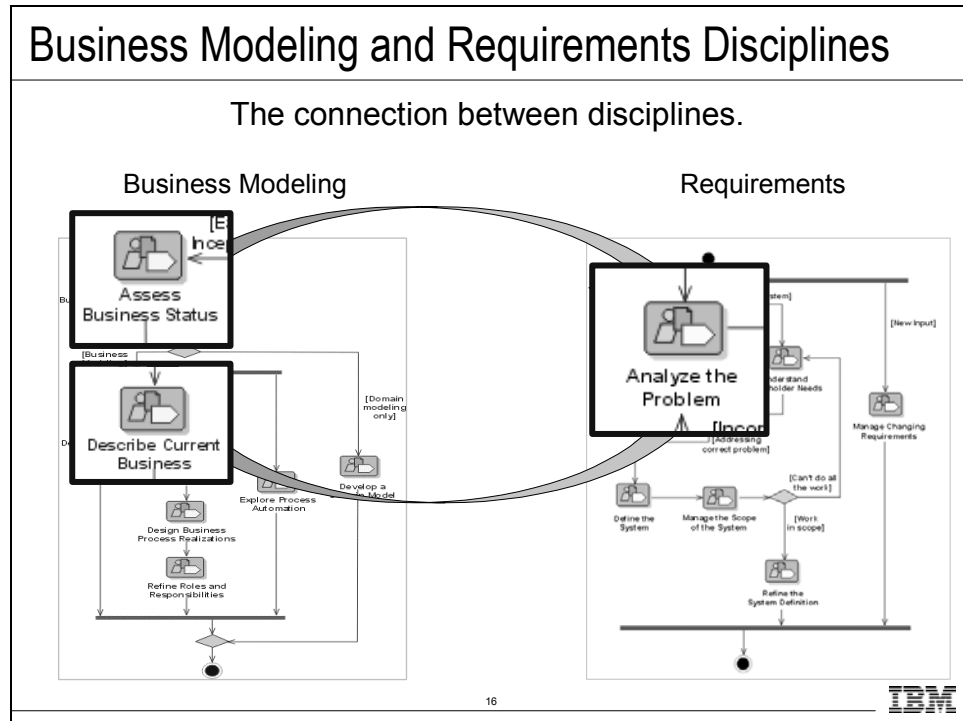
Some reasons to perform business modeling are:

- To identify what to automate in a business.
- To show how a tool can be used in the business.
- As the first step in re-engineering the way a business currently conducts business.

Some reasons to skip business modeling are:

- The product is a commercial product, rather than a contract (such as MS Word).
- The product's domain is well understood by the development team and clients.

Business Modeling and Requirements Disciplines



To truly understand a problem you must understand the broader context of the environment the problem exists within. Business modeling is a tool you can use to help you understand the bigger picture. The Rational Unified Process provides a discipline that guides you through the business modeling activities.

Business Models

Business Models

- ◆ Model organization structure and dynamics.
 - Business processes
 - Organizational structure
 - Roles and responsibilities
 - Products
 - Deliveries
 - Events
- ◆ Visualize the organization and its processes.
- ◆ Help understand current problems.
- ◆ Identify potential improvements.
- ◆ Derive and validate system requirements needed to support the organization.

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Business models enable us to consider the broader context. Even when we are focused on a specific application to be implemented, a business model enables us to develop the correct solution that solves the business problem.

The purposes of business modeling are to:

- Understand current problems in the target organization and identify improvement potentials.
- Understand the structure and the dynamics of the organization in which a system is to be deployed (the target organization). The business model provides a visual and textual model of the organization and its processes.
- Ensure that customers, end users, and developers have a common understanding of the target organization.
- Derive the system requirements needed to support the target organization.

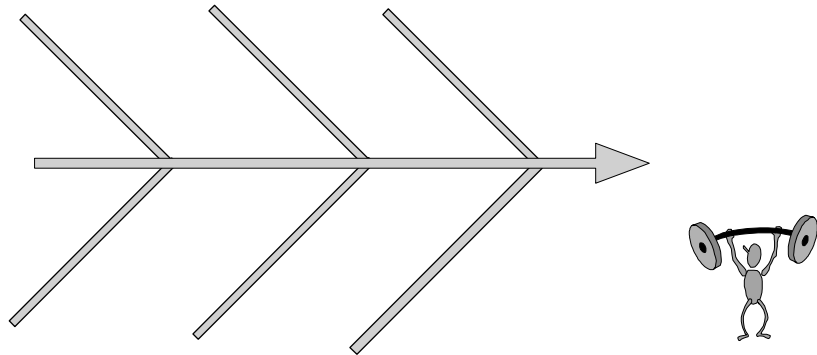
To achieve these goals, the business modeling describes a vision of the target organization. Based on this vision, the business model defines the processes, roles, and responsibilities of that organization.

If a business model is available, it can provide important input into the requirements elicitation process. It provides a context for system requirements, and it promotes a common understanding of the needs of the target organization(s).

Exercise 4.1: Identify the Problem

Exercise 4.1: Identify the Problem

- ♦ Discuss the class project.
- ♦ Identify and rank root causes.
 - Fishbone diagram

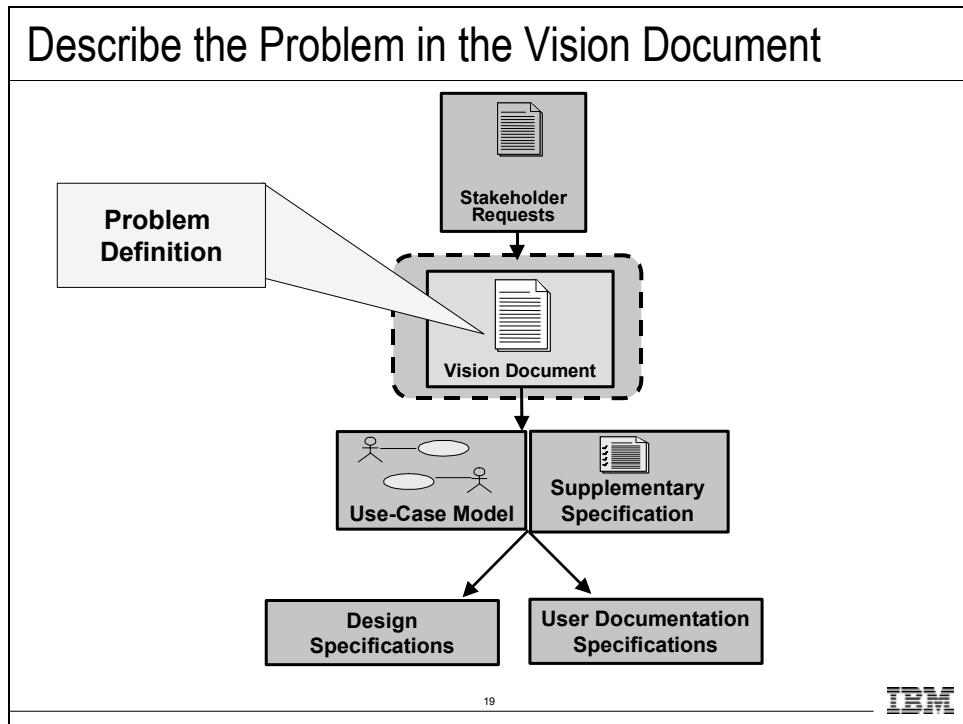


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Now that we've discussed how to analyze a problem, it is time to try doing it.
See Student Workbook Exercise 4.1: Problem Behind the Problem.

Describe the Problem in the Vision Document



Here is the requirement structure you use in the class and the specification artifacts in which they are captured. In this module, you focus on defining the Vision for the product.

Vision Document

Vision Document

- ♦ Communicates information between management, marketing, and the project team.
- ♦ Provides initial customer feedback.
- ♦ Fosters general understanding of the product.
- ♦ Establishes scope and priority of high-level stakeholder requests and features.
- ♦ A system-level document that describes the “what” and “why” of the product.

A document that gets “all parties working from the same book.”



Vision

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The primary purpose of any document is communication. You should always be aware of who is reading the document and ensure that they understand its content.

The Vision document provides a high-level (sometimes contractual) basis for the more detailed technical requirements. There can also be a formal requirements specification. The Vision captures very high-level requirements (features) and design constraints. It gives the reader an understanding of the system to be developed. It provides input to the project-approval process and is, therefore, intimately related to the Business Case. It communicates the fundamental “whys and whats” related to the project, and it is a gauge against which all future decisions should be validated.

In general, the Vision document is read by managers, funding authorities, use-case modelers, and developers.

The Vision is a system-level document that describes the “what” and “why” of the product or application.

Focuses on:

- Describing the problem
- Goals and objectives of the system
- Identifying and describing the stakeholders
- User needs
- Target markets
- User environments and platforms
- Product features

Vision Document Outline

Vision Document Outline

1. Introduction
2. Positioning
3. Stakeholder and User Descriptions
4. Product Overview
5. Product Features
6. Constraints
7. Quality Ranges
8. Precedence and Priority
9. Other Product Requirements
10. Documentation Requirements
11. Appendix 1 - Feature Attributes

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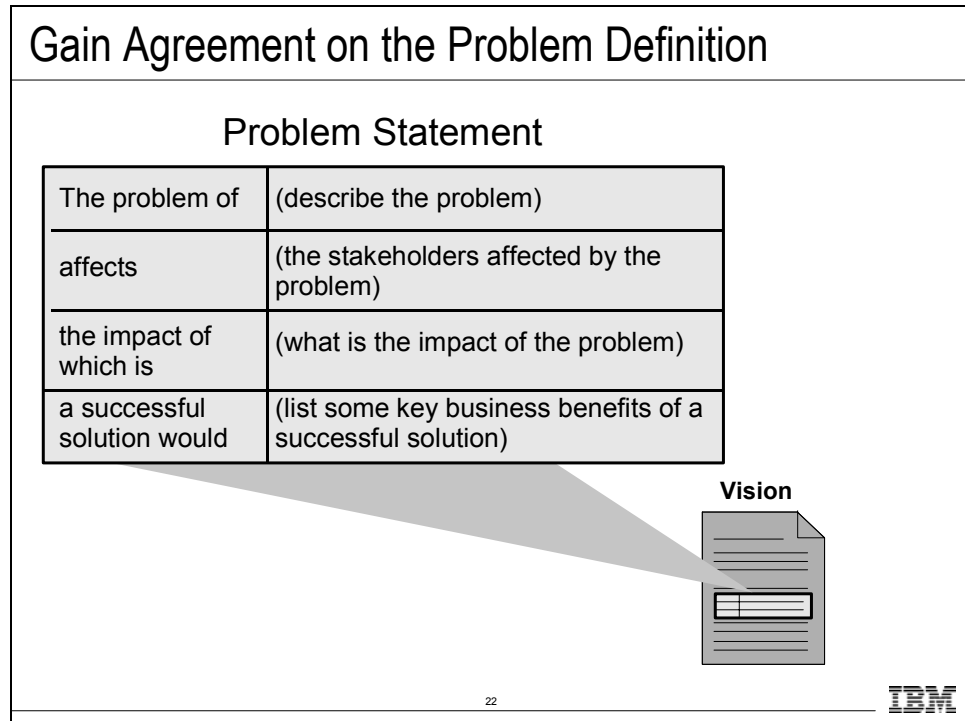
The main activity in the Analyze the Problem Workflow Detail is to develop the Vision for the system. The Vision contains a description of the problem and the high-level definition of the system.

This outline shows the major sections of the Vision document. As you can see, the Vision provides a wealth of information about the product to be built. In the *Develop the Vision* activity, make sure you understand and record the decisions made with your customer about all of the kinds of information in the defined by the Vision template.

This information is gathered together in constructing your Vision for the product to be built. This slide summarizes the major activities you have done and how they contribute to the Vision.

A sample Vision document can be found in the Student Handouts.

Gain Agreement on the Problem Definition



It is often useful to have the group agree on a short, clear statement of the agreed-upon problem that our product addresses.

Example: (for a customer support system)

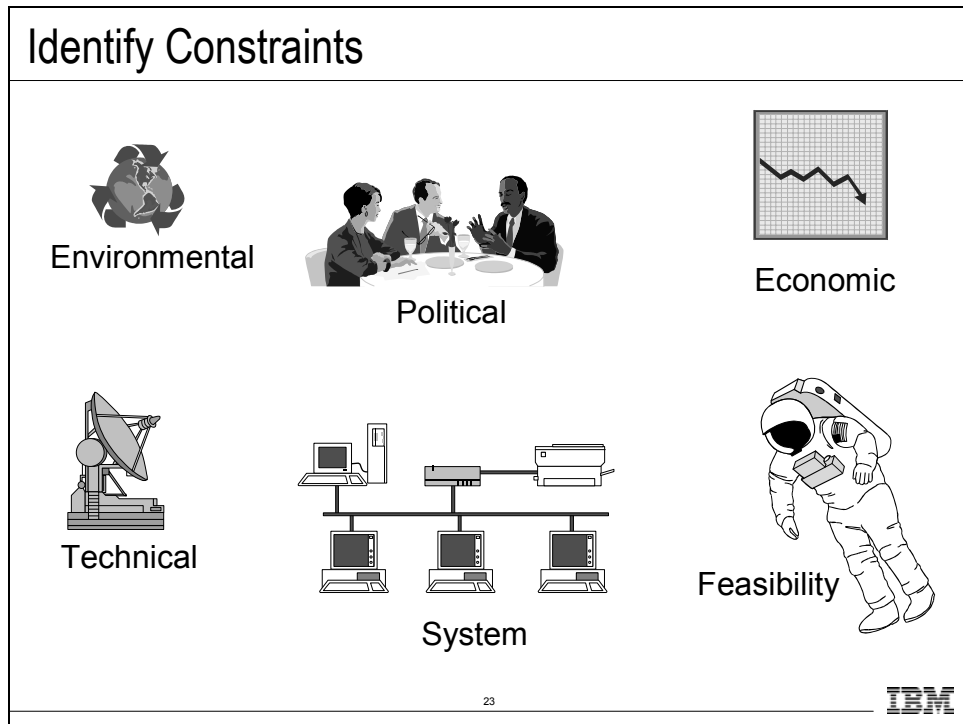
The problem of untimely and improper resolution of customer service issues

affects our customers, customer support representatives, and service technicians.

The impact of which is customer dissatisfaction, perceived lack of quality, unhappy employees and loss of revenue.

A successful solution would provide real-time access to a trouble-shooting database by support representatives, and facilitate dispatch of service technicians, in a timely manner, only to those locations that genuinely need their assistance.

Identify Constraints



Constraints were defined in the Introduction to MRMUC module.

Project constraints have the potential to severely impact the ability to successfully provide a solution. Sometimes a constraint may loom like an iceberg. On the surface, there may not appear to be much to it, but on close examination the iceberg may have a large potential impact. Maybe it will not be enough to sink the ship, but it may be enough to knock you way off course!

The types of constraints you identify may also influence our choice of elicitation techniques. This is particularly true of political constraints.

What types of constraints have you experienced in your projects?

Identify the Best Business Solution

Identify the Best Business Solution

- ♦ Identify a number of solutions for the main contributing problems.
 - Technical, non-technical, or both.
- ♦ Choose the ones that:
 - Best solve the root causes.
 - Support the business' goals.
- ♦ Gather the requirements to implement the solution.

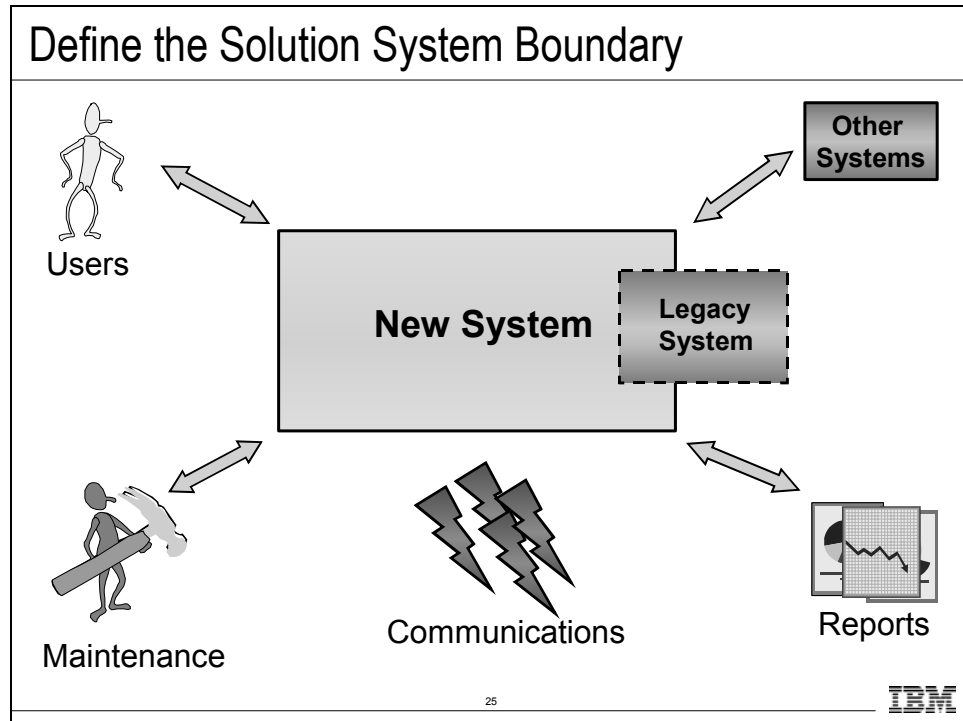
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Once you have all agreed on the problem, you should determine which solution is the best solution.

There may be more than one solution to a problem. The key is to come up with the best one.

Define the Solution System Boundary

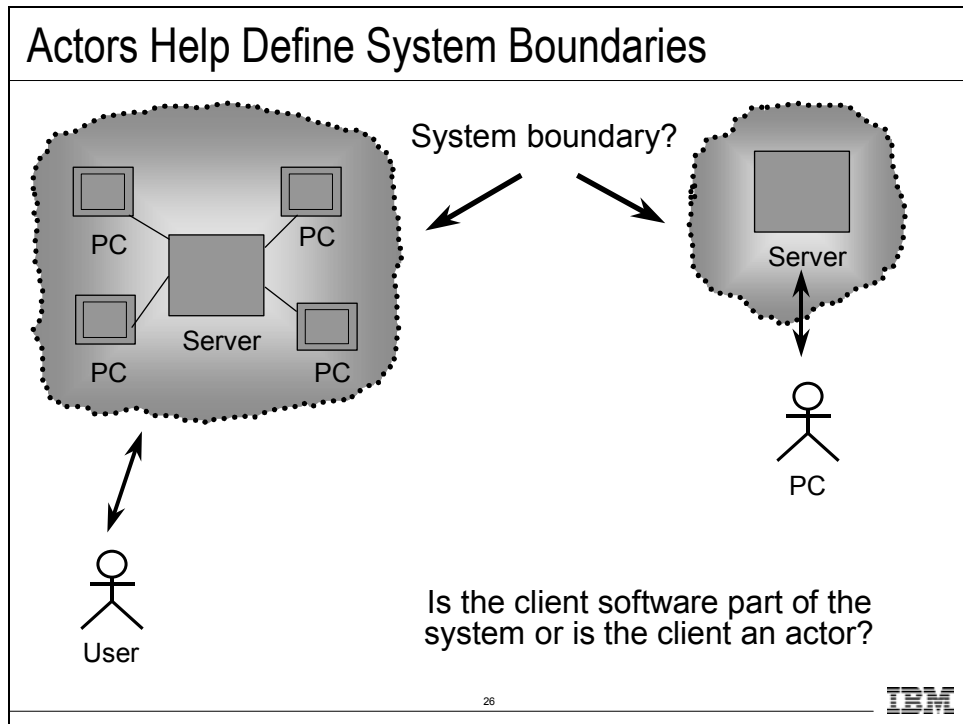


When you define the boundaries of a system you must consider what existing resources you can leverage (other and legacy systems). Who uses the system? Who maintains the system? Who receives the output of the system? (Reports.) How do you communicate with other systems? (Communications.) How do the capabilities of the new system overlap or replace any existing system?

What are the interfaces to outside people or systems for your project?

Based on our definition of actor, which of these are actors?

Actors Help Define System Boundaries



An **actor** is a user of the system but is not part of the system to be developed. A user can be a person or an external system.

Identifying the actors helps to define the boundaries of the problem to be solved. Because an actor is not part of the system to be built, you do not have to build the functionality that an actor represents. So, by identifying the actors, you help identify the boundary of the system to be built.

If we are developing the client software for the PCs, then the *User* would be the actor. If we are developing only the server software, then the *PC* is the actor.

Capture a Common Vocabulary

Capture a Common Vocabulary

- ◆ Define terms used in the project.
- ◆ Help prevent misunderstandings.



Glossary

Capture common vocabulary

- Start as soon as possible.
- Continue throughout the project.



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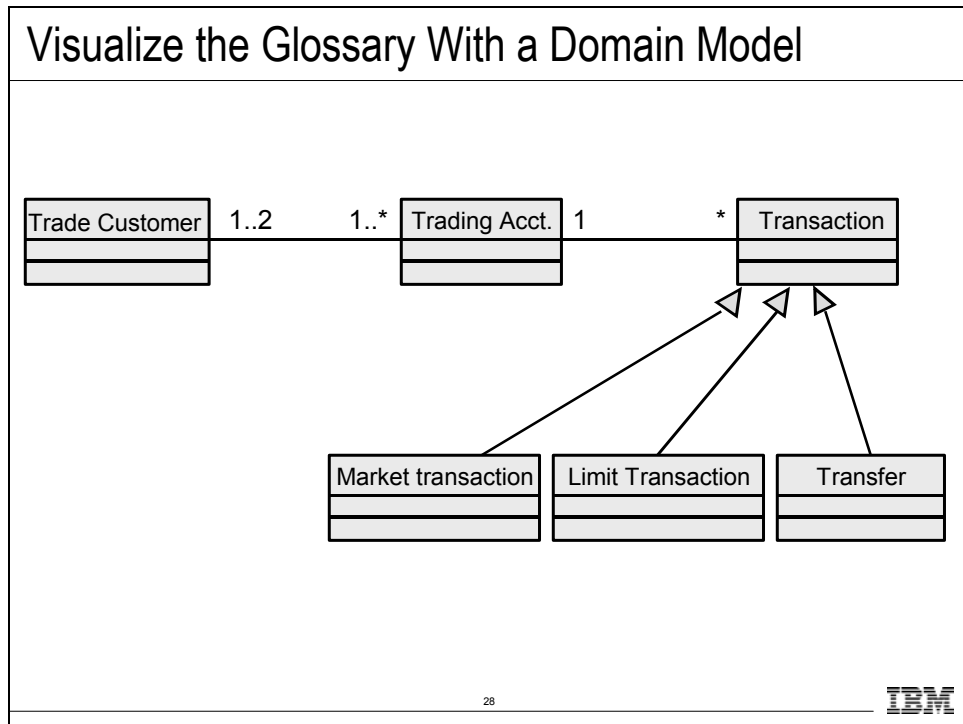
Another important part of beginning a project is to start identifying the common vocabulary. Look at the Glossary for a sample.

There should be one Glossary for a system. This document is important to all stakeholders, especially when they need to understand and use terms that are specific to the project. There is a sample glossary RUCS2 section of the Student Workbook.

All textual descriptions of the system, especially use-case descriptions should use and refer to the terms defined in the glossary.

In some cases, you may build a domain model to further visualize the terms in the glossary.

Visualize the Glossary With a Domain Model



A domain model provides a visual model of key business objects that are captured in the glossary. The benefit of a domain model is that the relationships between the glossary terms can be represented visually, thereby improving the stakeholders domain understanding.

This is an example of part of a domain model for the RU e-st system. It shows that there are trading customers, trading accounts, and trading transactions. There are three types of trading transactions: market transactions (buy/sell at the current market price), limit transactions (buy/sell at a specific price), or transfers between trading accounts.

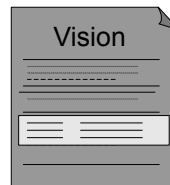
The numbers on associations between objects indicate how many objects a particular object can be related to. This diagram shows:

- One trading customer must have one account and can have an unlimited number of accounts.
- One account is owned by either one trading customer (a single account owner) or two trading customers (a joint account).
- One transaction (for example, one order to buy stock) is associated with a single account.
- One trading account can have an unlimited number of transactions associated with it.

Exercise 4.2: Describe the Problem

Exercise 4.2: Describe the Problem

- ◆ Start the Vision document.
 - Identify project stakeholders.
 - Find actors and system boundaries.
 - Identify constraints on the project.
 - Formulate a problem statement.



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See Student Workbook Exercise 4.2: Describe the Problem.

The purpose of this exercise is to practice describing the problem in the Vision document.

Review: Analyze the Problem

Review: Analyze the Problem

1. What are the activities in problem analysis?
2. How do you gain agreement on the problem?
3. Who are the stakeholders in your project?
4. How can actors be used to help determine the boundaries of a system?
5. Why is it important to establish a glossary?
6. What should be included in a problem statement?

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1.

2.

3.

4.

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